

Department of Electrical, Electronic and Computer Engineering

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Design and Implementation of a Secure Windows Server 2019 Infrastructure for Enterprise Network Services

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by

Group 1

Group project report submitted in partial fulfilment of the requirements for the degree

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in the Faculty of Engineering and the Built Environment

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AI Declaration

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- **LaTeX formatting assistance** (structure and code improvements).

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Abstract

This report presents the installation, configuration, and testing of a Windows Server 2019 environment developed to meet the network and security requirements of CPUT. The project was implemented using the hypervisor utilised during Semester 1 and included the deployment of several key server roles and services.

The implementation involved configuring Active Directory Domain Services (AD DS) for centralized user and resource management, Domain Name System (DNS) for name resolution, and Dynamic Host Configuration Protocol (DHCP) for automatic IP address allocation. In addition, Group Policy settings were applied to improve the security of the environment by restricting Command Prompt access, disabling removable storage devices, preventing unauthorized software installations, and enforcing stronger password policies.

Remote Access Services and Windows Certificate Services were also configured to provide secure connectivity and authentication within the network environment. Each service was tested to verify that it functioned correctly and met the specified requirements. Screenshots of all installation, configuration, and testing procedures were captured and included as evidence of successful implementation.

The results demonstrate how Windows Server 2019 can be used to create a secure and manageable network infrastructure that supports organisational requirements while following recommended security practices.

Keywords: CPUT, Windows Server 2019, Active Directory, DNS, DHCP, Group Policy, Remote Access, Certificate Services, Network Administration, Security Policies, Server Configuration

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1. Introduction

As organisations continue to grow, the need for a reliable and secure network infrastructure becomes increasingly important. CPUT requires an IT environment that can support its expanding operations while ensuring efficient management of users, devices, and network resources. To meet these requirements, a Windows Server 2019 environment was implemented using the hypervisor used during Semester 1.

This project focuses on the installation, configuration, and testing of key Windows Server services, including Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services. Screenshots were captured throughout the process to provide evidence of the installation and configuration steps performed.

1.1 Background

Windows Server 2019 is widely used in organisations to manage network resources, user accounts, security policies, and shared services from a central location. Services such as Active Directory, DNS, and DHCP help administrators manage users and devices more efficiently while ensuring that network communication functions correctly.

Security is also an important part of network administration. For this reason, several security policies were implemented in this project. These include disabling Command Prompt access, preventing the use of removable storage devices, restricting software installations, enforcing a minimum password length of ten characters, and requiring users to change their passwords regularly. These measures help reduce security risks and improve the overall protection of the network environment.

1.2 Objectives

The objectives of this report are:

- To install and configure Windows Server 2019.
- To configure and test Active Directory Domain Services (AD DS).

- To configure and test DHCP services.
- To configure and test DNS services.
- To implement Group Policy settings that improve system security.
- To configure Remote Access services.
- To install and configure Windows Certificate Services.
- To verify that all configured services are functioning correctly.
- To document all installation, configuration, and testing procedures using screenshots.

1.3 Report Structure

The remainder of this report is organised as follows:

- Chapter 2 covers the installation and initial configuration of Windows Server 2019.
- Chapter 3 discusses the configuration and testing of Active Directory Domain Services (AD DS).
- Chapter 4 explains the configuration and testing of DHCP services.
- Chapter 5 presents the configuration and testing of DNS services.
- Chapter 6 describes the implementation of Group Policy settings.
- Chapter 7 discusses the configuration of Remote Access services.
- Chapter 8 explains the installation and configuration of Windows Certificate Services.
- Chapter 9 presents the testing and verification of the configured services.
- Chapter 10 provides the conclusion of the project.
- The final section contains the references used in this report.

2. Installation and Initial Configuration of Windows Server 2019

2.1 Introduction

This chapter describes the installation and initial configuration of Windows Server 2019 within the virtualised environment used during Semester 1. The purpose of this stage was to establish a functioning server that could support the deployment of the network services required for the project.

The installation process involved preparing the virtual machine using VirtualBox, installing the operating system, and completing the initial server configuration. These steps provided the foundation for the subsequent configuration of Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services.

2.2 Virtual Machine Preparation

Before installing Windows Server 2019, a virtual machine was prepared using VirtualBox. Virtualisation allows multiple operating systems to run on a single physical computer while maintaining separate virtual environments.

The virtual machine was configured with the necessary resources to support the operating system and the planned server roles. These resources included virtual memory, processor allocation, virtual storage, and a network adapter. The Windows Server 2019 installation media was then attached to the virtual machine, allowing it to boot from the installation image.

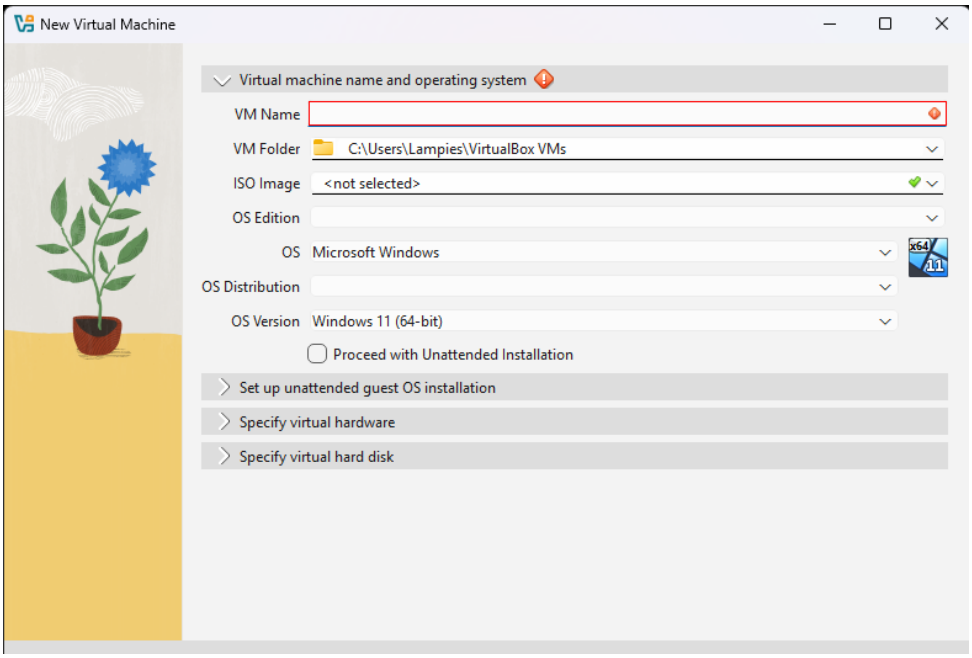


Figure 2.1: Initial Virtual Machine Configuration.

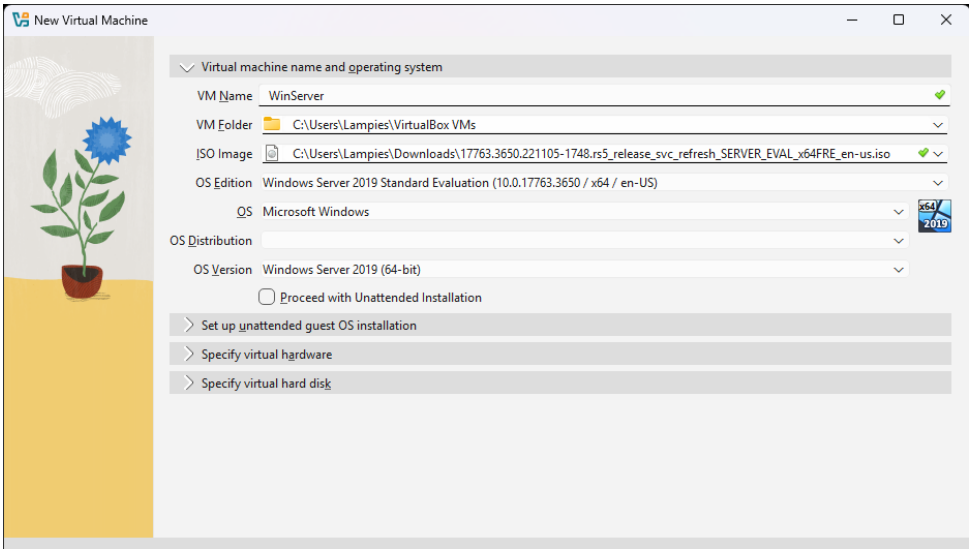


Figure 2.2: Virtual Machine Configuration Step 2.

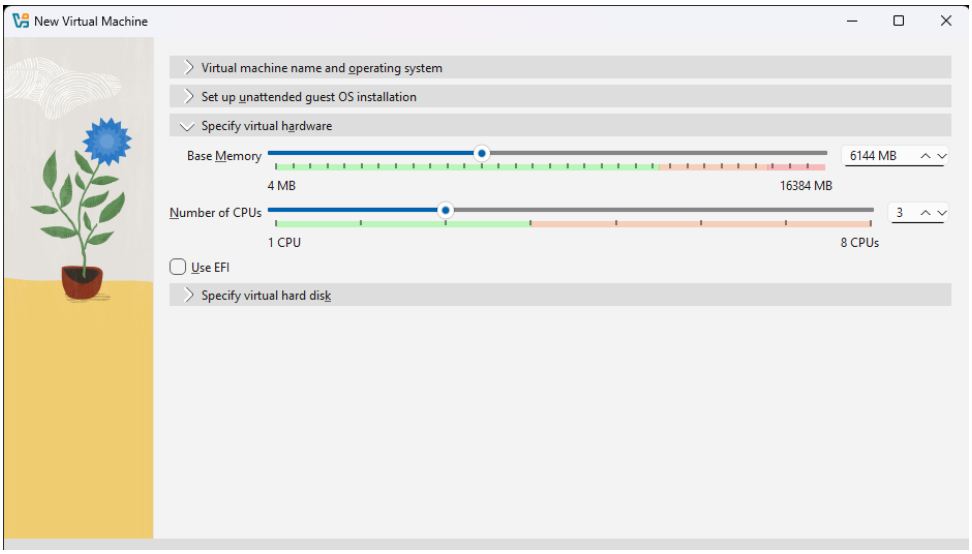


Figure 2.3: Virtual Machine Configuration Step 3.

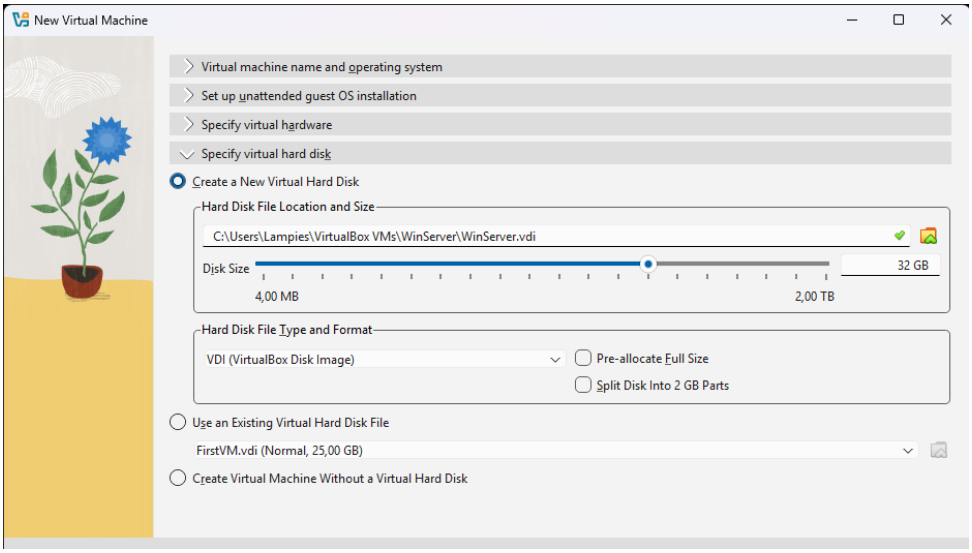


Figure 2.4: Virtual Machine Configuration Step 4.

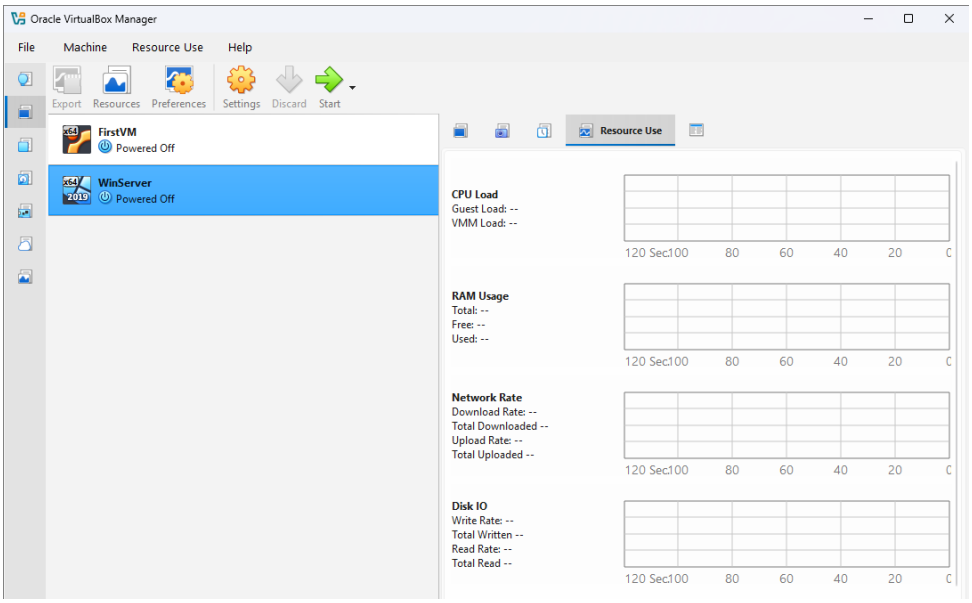


Figure 2.5: Virtual Machine Configuration Step 5.

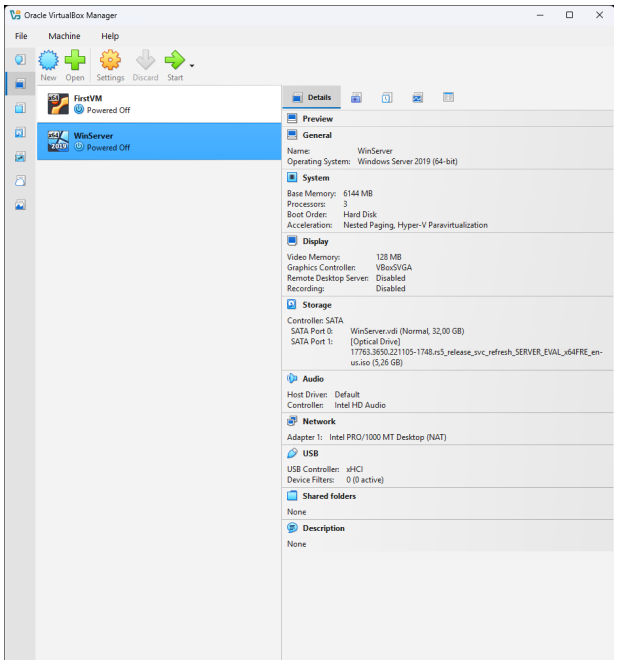


Figure 2.6: Final Virtual Machine Configuration Step.

2.3 Installation of Windows Server 2019

Once the virtual machine had been configured, it was started and the Windows Server 2019 installation process was initiated. The installation wizard was used to select the appropriate language, time and currency format, and keyboard layout.

The required Windows Server 2019 edition was then selected. The installation proceeded by accepting the applicable licence terms, selecting the installation type,

and choosing the virtual disk on which the operating system would be installed.

After the installation files had been copied and the required setup operations completed, the virtual machine restarted. The initial administrator password was configured, allowing access to the server using the local Administrator account.

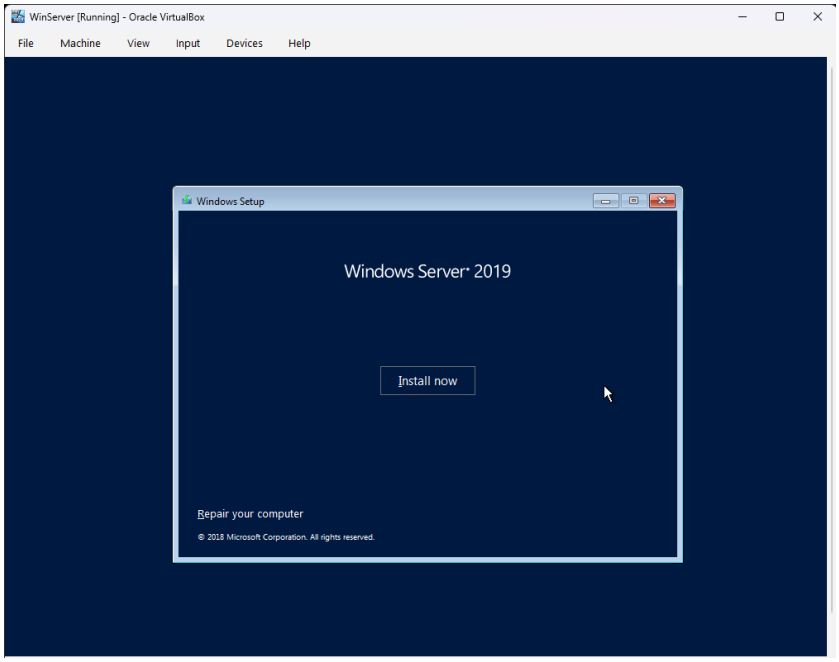


Figure 2.7: Installation Screen 1.

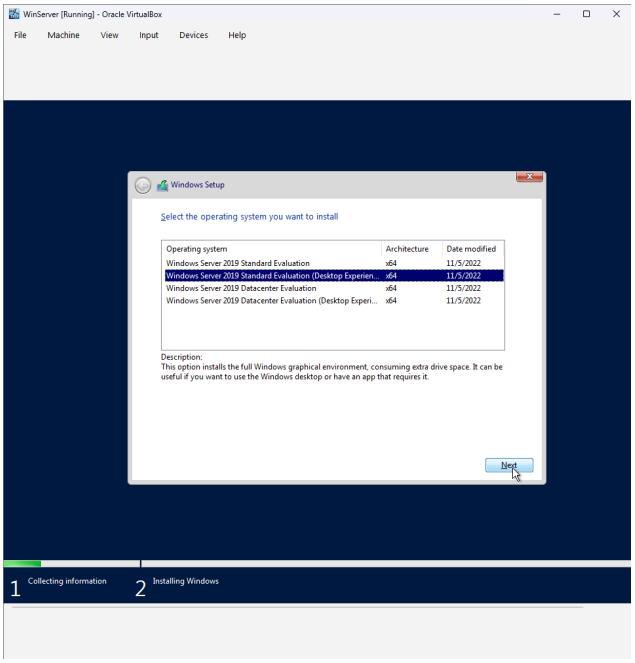


Figure 2.8: Installation Screen 2.

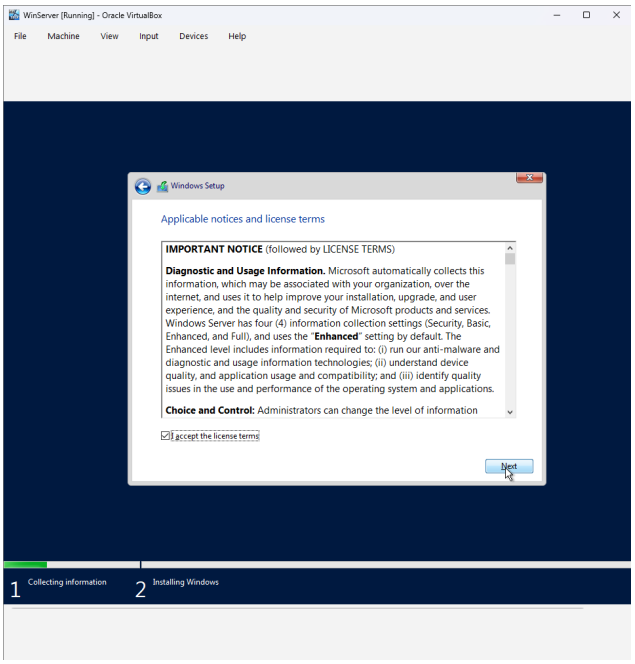


Figure 2.9: Installation Screen 3.

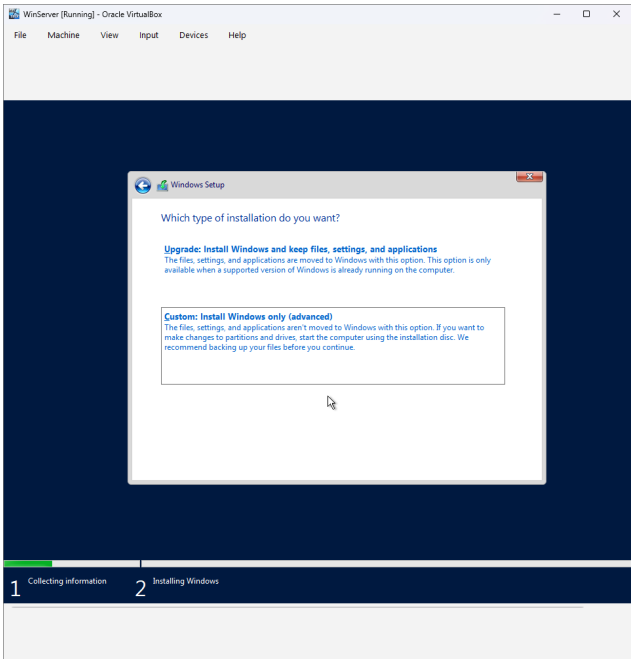


Figure 2.10: Installation Screen 4.

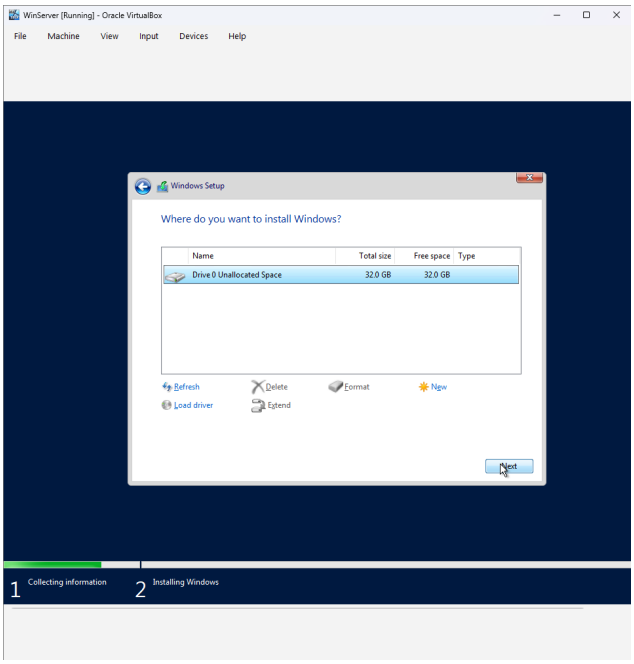


Figure 2.11: Installation Screen 5.

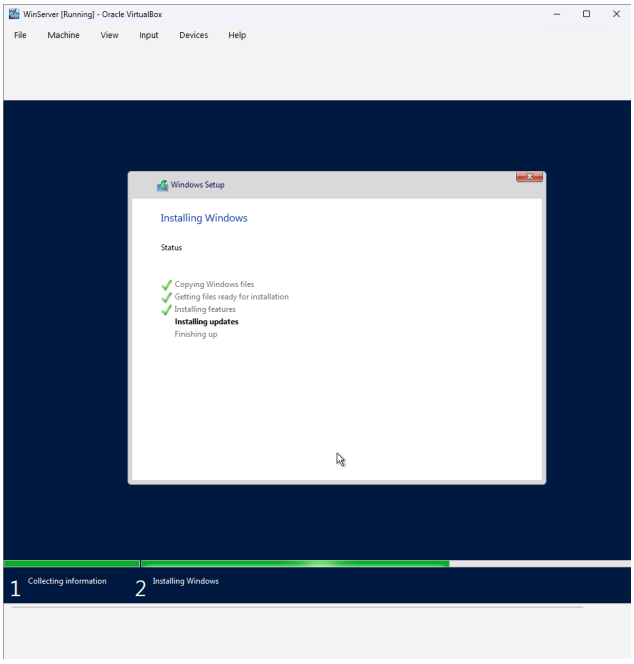


Figure 2.12: Installation Screen 6.

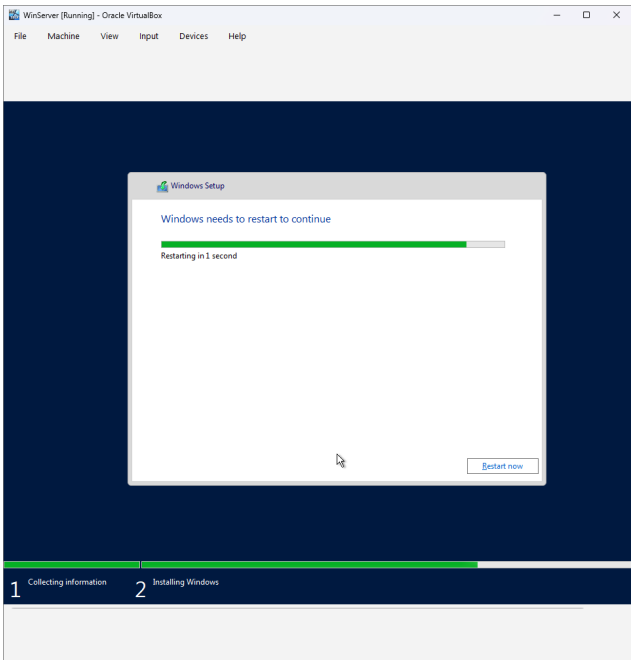


Figure 2.13: Installation Screen 7.

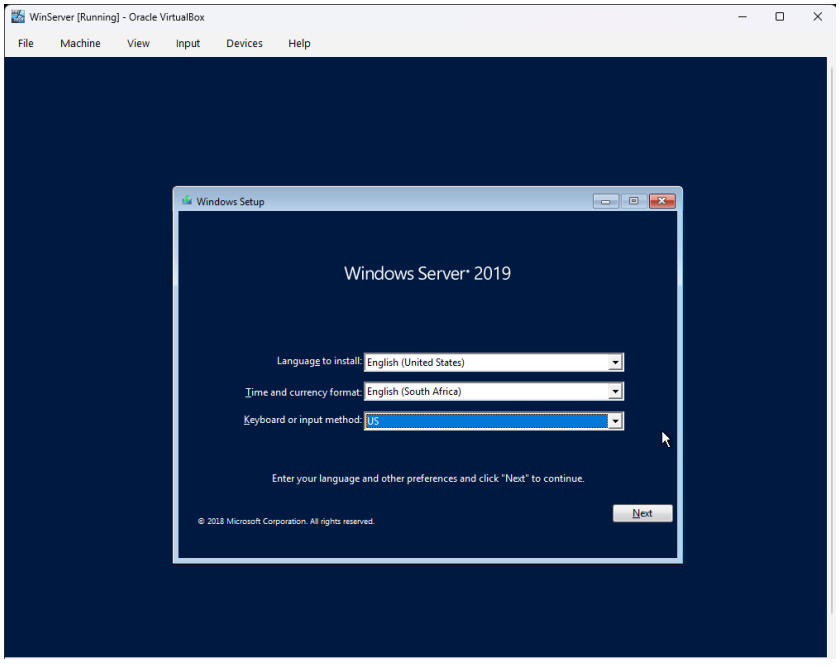


Figure 2.14: Installation Screen 8.

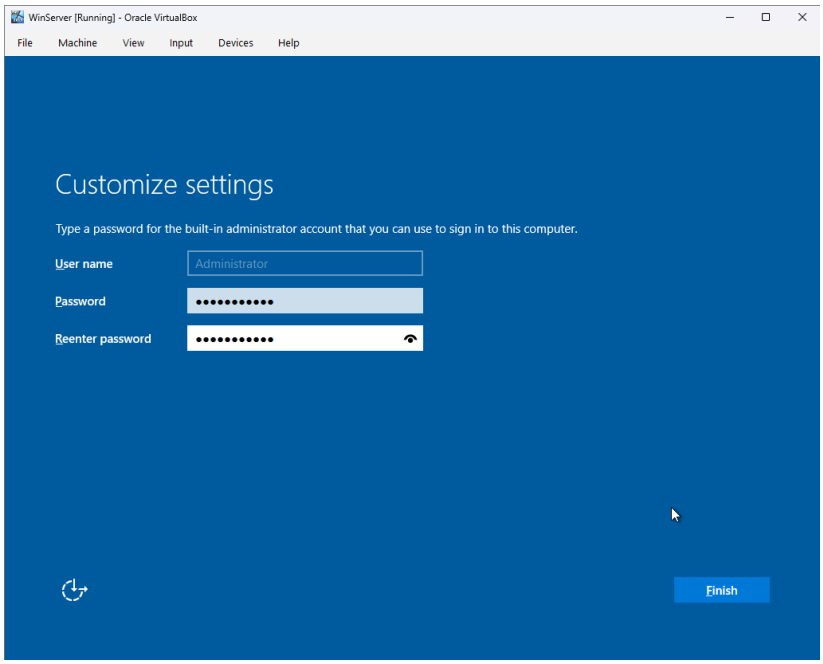


Figure 2.15: Windows Server Account Set-Up.

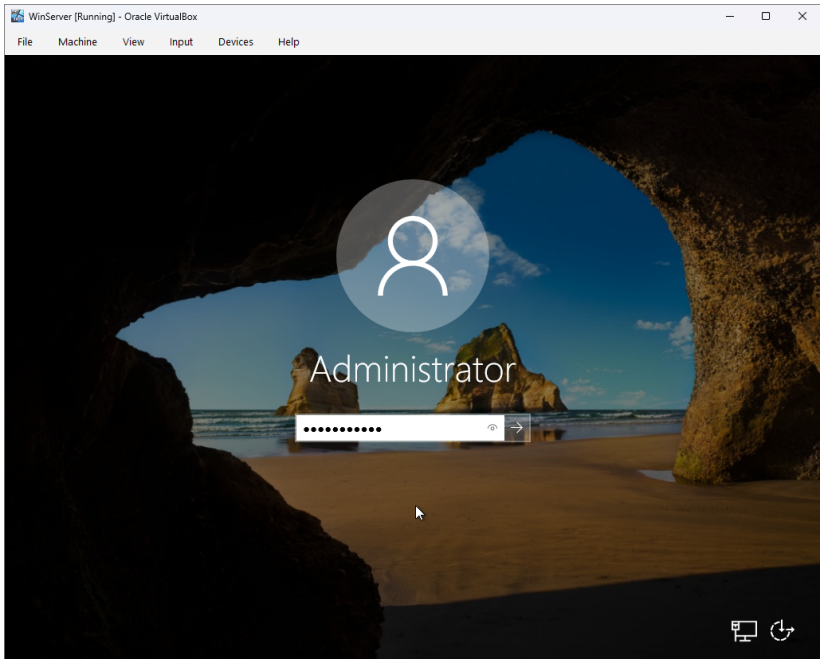


Figure 2.16: Windows Server Log-In Screen.

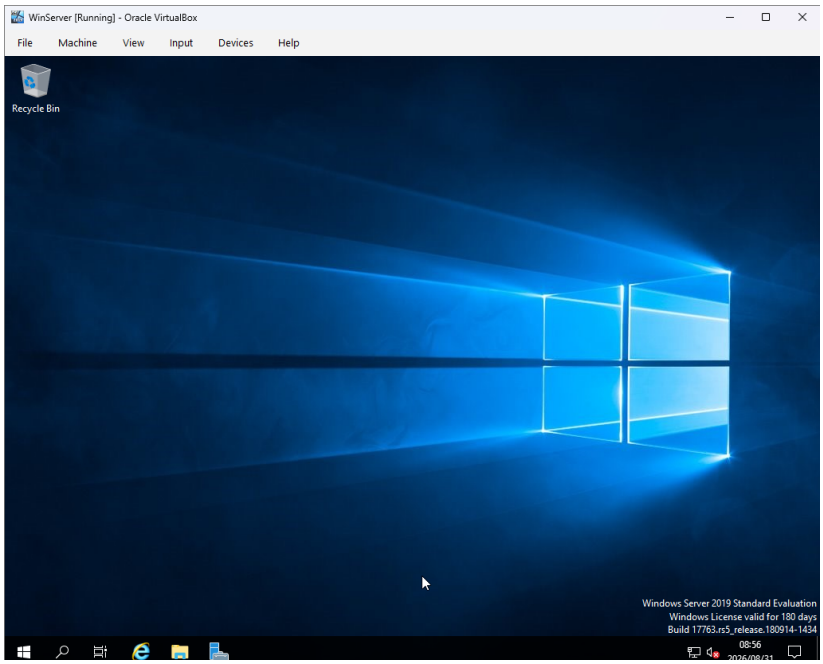


Figure 2.17: Windows Server Loaded and installed.

2.4 Testing

Testing was conducted to verify that Windows Server 2019 had been installed successfully and that the initial server configuration was functioning as expected. The tests focused on operating system availability, server identification, network configuration, and basic network connectivity.

2.4.1 Operating System Verification

The server was restarted to confirm that Windows Server 2019 could boot successfully after installation. The administrator was able to log in, and the Server Manager dashboard was accessible. This confirmed that the operating system was operational and ready for further configuration.

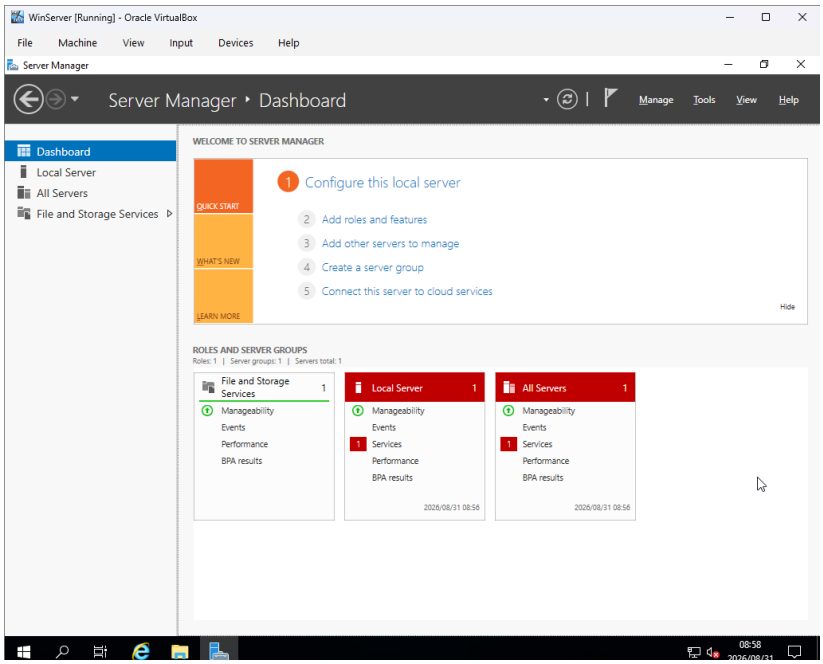


Figure 2.18: Server Manager dashboard following the installation of Windows Server 2019.

The dashboard provided access to the server's management tools and the interface used to install and configure additional server roles and features.

Further testing should verify the server hostname, IP configuration, and network connectivity. The following commands can be used to perform these checks:

- `hostname` – displays the configured computer name.
- `ipconfig /all` – displays the network adapter configuration.
- `ping <gateway-IP>` – tests connectivity to the configured network gateway, where applicable.

The outputs should be recorded and included as supporting evidence. Any failed tests should be investigated and resolved before proceeding with the configuration of additional server roles.

More comprehensive testing of Active Directory Domain Services, DNS, DHCP, Remote Access, and Windows Certificate Services is addressed in the relevant chapters of this report.

2.5 Chapter Summary

This chapter described the preparation of the virtual machine, installation of Windows Server 2019, and completion of the initial server configuration. The process established the operating system environment required for the deployment of additional server roles and network services.

Initial verification procedures were also described to assess operating system availability, server identification, network configuration, and connectivity. These steps establish the foundation for the remaining stages of the project.

The next chapter discusses the configuration and testing of Active Directory Domain Services (AD DS), which provides centralised management of users, computers, and authentication within the network environment.

3. Conclusion

3.1 Summary

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3.2 Key Findings

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3.3 Future Work

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3.4 Final Remarks

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A. Appendix

A.1 Github Repository